

Integration of Concept Maps into the Mixed Reality Learning Space: Quasi-Experimental Design and Preliminary Results

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ABSTRACT

It is difficult to establish the relationship among complex concepts learned in virtual learning activities in the absence of appropriate instructional scaffolds, which may lead to students' confusion and frustration. To address such an issue, a mixed reality (MR) friction experiment system integrated with interactive hierarchical concept maps with the name of MMRCM is proposed as a scaffolding tool to guide students in learning friction concepts. Results of a quasi-experiment assessing the effectiveness of MMRCM showed that students' academic performance improved significantly. Interviews with students and teachers also showed that the MMRCM system could help learners navigate educational concepts and material efficiently.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Visualization—Visualization application domains—Scientific visualization

1 INTRODUCTION

Visualization is one of the most critical technologies applied in the field of education, which can arrange detailed information into an interface in a compact way and has been studied extensively. In MR learning applications, abstract learning contents can be visualized into interactive virtual objects, which reduces the complexity of presenting intangible phenomena and makes it easier for learners to acquire knowledge of the underlying concepts. However, if the MR learning materials lack appropriate learning structure and instructional scaffolds, learners will encounter difficulties in knowledge exploration. Meanwhile, such graphical tools as concept maps are used to plan curriculums or instruction on a specific topic, making the instruction "conceptually transparent" to students [1]. Although MR and concept maps have both proved to be promising visualization methods to support meaningful knowledge organization for learners with limited cognitive abilities, few studies have investigated whether combining these two powerful visualization tools can be beneficial to learners in experimental learning. To fill such gap, this paper proposed a mobile MR learning system using concept maps (MMRCM) as a knowledge navigation interface, aiming to explore the pedagogical affordance of combining these two visualization techniques. A quasi-experiment was conducted to investigate the effectiveness of MMRCM as well as the perceptions of teachers and students.

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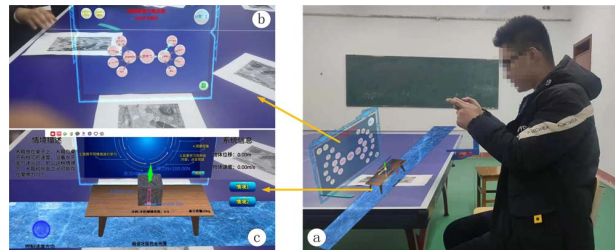


Figure 1: The overview of the MMRCM system. (a) one student conducts the experiment using the MMRCM. (b) The view of the concept maps in three-dimensional space. (c) The MR learning scenario hyperlinked with the concept map.

2 METHODOLOGY

In this section, we describe the process of designing and implementing MMRCM as well as the evaluation procedures of the quasi-experiment.

2.1 System Design

The proposed MMRCM system design consists of two main blocks: concept map construction and MR learning space construction.

In this study, the concept map is constructed as a visual representation of abstract concepts according to the WYSIWYG paradigm [2], where key concepts are represented as tagged nodes and the relationships among them are represented as tagged links. We presented the concept map in hierarchical radial layouts, which makes the design concise to display dependency relationships of concepts, as shown in Fig. 1. For example, as the fundamental concept, the friction is positioned at the concept map center. In the outermost layer, sub-concepts describing the frictional attributes are provided at a fine granularity. The concept nodes in the outermost layer are essential because these child nodes are the connections between the concept map and the MR learning space. In total, 14 sub-concepts in the outermost hierarchy were generalized from the syllabus.

Moreover, interactivity is one of the most striking characteristics of the constructed concept maps. The visual representation of the concept node is a 3D sphere that students can interact with by clicking. Students can collapse or expand parts of the concept map hierarchy through the clicking interaction with the node sphere, making the interface more compact or display more information. Appropriate visual guidance, such as visual cues or textual annotations, is needed to help students understand the concept map structure. In this study, the luminous effect of node spheres and the color change effect of tagged links were used as visual cues to guide students during concept mapping usage. We divided the learning process into three stages: pre-hyperlink, hyperlinking, and post-hyperlink.

-The pre-hyperlink stage. The node sphere that is going to be activated has the red glow effect as visual prompts.

-The hyperlinking stage. The activating node sphere has the yellow glow effect as visual prompts, and the arrow color on the path from the center node (friction) to the activating node changes from gray to yellow

-The post-hyperlink stage. The arrow color on the path from the center node(friction) to the unlocked node changes to blue, and at the same time, the next concept node in the outermost hierarchy goes into the pre-hyperlink stage.

After the concept map construction, the MR learning space was developed with substantial effort. The number of learning scenarios is equal to the number of nodes in the outermost hierarchy of the concept map, and the learning content of every learning scenario represents the corresponding sub-concept. The variable-controlling approach is adopted to make the sub-concept studied as the single variable in MR learning scenarios to prevent interference from other concepts. The force state and motion state of the virtual object are visualized with MR technology in real time. Moreover, students can manipulate virtual objects with the touch screen of mobile devices. So, students can learn the abstract concept about friction properties through the learning-by-doing approach and dynamic visualization of abstract force.

According to the spatial contiguity principle in Mayer's cognitive theory of multimedia learning, learning is enhanced when the spatial distance between related information elements is minimized. As a result, we put the concept map in the same 3D space as the MR learning scenario, making the concept map like the head-up display. Students can achieve a seamless transition between concept maps and MR learning scenarios.

2.2 System Implementation

We implemented the visualization representation of concept maps and MR learning scenarios using the popular game engine Unity 3D engine, and the Google ARCore [3] is used to track the pose of the navigation system. With the motion tracking capability of ARCore, students can use mobile devices to switch the viewpoint between the concept map and the learning scene without worrying about losing the target if the perspective is separated from the image marker.

2.3 Evaluation Procedures

An academic achievement questionnaire consisting of 20 items in which 15 items were modified from the FFCI (proposed in [4]) and 5 items were selected from other reference books based on friction characteristics is developed, all of which were assessed for expert validity by experienced physics teachers. Moreover, an extended technology acceptance model (TAM) was constructed to test whether students accepted MMRCM; however, the detailed analysis results of TAM are not included in this paper.

In order to explore the effect of MMRCM on students' academic performance, a quasi-experimental method was adopted, and the "one-group pre-test post-test design" method was adopted as the research model. Fifty-five high school students (19 females and 36 males) were invited to participate in this experiment, whose ages range from 17 to 19. This study was conducted in a two-week physics experiment class to learn the friction concepts in the autumn semester of 2020. Before the experiment, the students were given a tutorial about how to use the MMRCM. Then the students were required to finish the academic achievement questionnaire as the pre-test. Students use MMRCM as a scaffolding tool to learn the concept of friction over the course of four lessons, and they need to alternate experiments due to the limited number of devices. After every student has finished the experiment, a post-test was conducted using the same academic achievement questionnaire. In the end, five students and three physics teachers were invited to an interview about their perception of MMRCM.

3 EXPERIMENTAL RESULTS

The paired t-test was used in this study to determine whether the score difference between pre-test and post-test was significant. Table 1 presents the paired t-test results of the academic performance.

The analysis result shows that $t = 12.535$ and $p < 0.05$, which indicates that there is a significant difference in favor of the post-test between the academic scores. Therefore, it could be suggested that the MMRCM had a significant positive impact on high school students' concept learning ability.

Table 1: the paired t-test results of the pre-test and post-test scores

Test	N	Mean	Std. dev.	t(54)	p
Post-test	55	66.818	10.816	12.535	p<0.001
Pre-test	55	48.545	11.493		

$p < 0.05$.

Substantial valuable opinions and feedback were concluded through the interview with teachers and students after the experiment. Interviewees stated that they could understand the knowledge structure from a holistic perspective and have a clear understanding of which sub-concepts are difficult. For example, one of the students said that "concept maps can improve students' understanding and connection of knowledge." They also mentioned that it was fascinating and impressive to experiment with virtual objects in the MR environment. One teacher mentioned that "When students use the MR learning system to learn, the visual perception of MR will deepen their understanding of the perception, and the hands-on experience will play a very good role in the mastery of the perception."

4 CONCLUSION

In this paper, a novel MR learning system (MMRCM) is proposed, which integrates the interactive hierarchical concept map in the MR learning space. The experimental results show that MMRCM is a scaffolding tool suitable for guiding students to learn abstract concepts in experimental classes. The interviews also show that students and teachers have a high enthusiasm for this work. This study has certain enlightening significance for other researchers in knowledge navigation and scientific visualization in the virtual learning space.

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